Type MS · Ethics of technology: privacy, inclusion and professionalism

Answer 1 of 5 · Question 4 part f · 6 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part f in this mark scheme extract

Number			
4(a)	Award up to **two** marks for a linked explanation such as: • 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1) • The simulation can be rerun with different values (1) to see how the outcomes will change (1) • The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1) • The simulation can be safer (1) because of the danger of animal interaction (1) • The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1)	Looking for a relevant statement and a linked extension/expansion. Do not award two independent statements that are not linked. Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ.	2
4(b)	**The only correct answer is C** ***A** is not correct because it is not a linear scenario* ***B** is not correct because parallel models do not have to communicate and coordinate their actions* ***D** is not correct because there is not an explicit input with an expected output*		1

Type MS · Ethics of technology: privacy, inclusion and professionalism

Answer 1 of 5 · Question 4 part f · 6 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part f in this mark scheme extract

4(c)	Award up to **two** marks for a linked description such as: • The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1) • Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1) • The research facility could implement a security question (1) that only the employee would know the answer to (1)	Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question. Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated	2
4(d)(i)	Award **one** mark for any of the following: • The manipulation of matter with a size from 1 to 100 nanometres / scale of 10^{-9} / at atomic/molecular level (1) • The inclusion of nanoparticles into different materials (1) • Facilitates miniaturisation at atomic/molecular **scale/level** (1)	Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level.	1
4(d)(ii)	Award **one** mark for any of the following: • Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1) • Miniaturisation of electronics e.g. transistors (1) • Nanomedicines to deliver drugs (1)	Allow examples that reasonably map to the categories given.	1

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Type MS · Ethics of technology: privacy, inclusion and professionalism

Answer 1 of 5 · Question 4 part f · 6 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part f in this mark scheme extract

4(e)	Award **one** mark for each of the following up to a maximum of **three** marks: • It uses a total of 128 bits (1) • There is a total of 32 hexadecimal character (1) • Each hexadecimal character requires 4 bits (1) • There are 8 blocks of 4 character / 8 x 4 blocks (1) • Each block (of 4 character) requires 16 bits (1) • The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1) • Each of the 8 blocks uses 16 bits / 8 x 16 (1)		3
4(f)	Indicative content • Rights and responsibilities • Data Protection regarding personal data • Surveillance • Consent of the individual to having specified data held • Accountability of those running / using such systems • Ownership • Data mining & analysis of work patterns and potential use/misuse • Ethics versus the legality of implementing such systems • Data misuse and protection against hacking and malware • Requirement to keep data secure • Physical access to areas of the site • Monitoring employee movement to restricted areas / resources • Algorithmic bias in facial recognition		6
		Total for question 4	**16**

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Answer 1 of 5 · Question 4 part f · 6 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part f in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 1 of 5 · Question 4 part f · 6 marks

2024 JUN · Paper 01 · extract 5 of 5 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Ethics of technology: privacy, inclusion and professionalism

Answer 2 of 5 · Question 6 part b · 2 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

6(a)(i)	Award **two** marks for a linked explanation such as: • Uses DNA/biomolecular components (1) rather that standard hardware/silicon chips (1) • DNA uses four-character genetic alphabet (1) rather than binary digits (1) • DNA uses chemical reactions (1) rather than electrical states/properties (1) • DNA computing uses massive parallel processing/strands (1) to solve problems that otherwise would take impossible amounts of time (1)		2
6(a)(ii)	Award **one** mark from: • The ability of a system to be in multiple states at the same time (1) • Multiple probabilities at the same time (1) • One state combines all possibilities (1) • The state is not known until it is measured (1)		1

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Answer 2 of 5 · Question 6 part b · 2 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **two** marks from: • Some people may not want to use technology (1) • Some people may not be able to afford to use technology (1) • Some people may not have access to the technology/infrastructure (1) • Some people may not be allowed access to technology (1) • Some people may not be able to use it (disabilities etc.) (1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)	Indicative content Impacts • Pollution (water, air, noise) resulting from the manufacturing process • High energy and water volumes needed for the manufacturing process • Mineral mining contaminates ground water • Mineral (copper, gold, silver, lithium) resource depletion • High energy use to keep machines cool with air conditioning or fans • Contain toxic components which means computing devices are hazardous waste • Batteries (laptop, lithium cells) disposal is hazardous and needs specialised disposal facilities • Computing devices sent to landfills contaminate ground and water resources Actions • Check national and local legislation • Some countries/states/regions require sellers to recycle own waste • Research recycling facilities to see if e-waste is acceptable • Find if local/national/regional government agencies charge for collection and disposal of e-waste • Find sellers offering exchange old for new + cost purchase options • Find charities (local, regional) which take donations of old devices • Turn off machines when not in use to save energy		6

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Answer 2 of 5 · Question 6 part b · 2 marks

2021 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content
Level 1	1-2	Basic independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

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Answer 2 of 5 · Question 6 part b · 2 marks

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Answer 3 of 5 · Question 5 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

5(a)(i)	Any **two** from: • It will interpret/analyse patient input to identify symptoms (1) and match the symptoms to (possible) illnesses (1) • It will match symptoms to possible illnesses (1) and give the most likely/probable illness (1) • It will match symptoms to possible illnesses (1) and ask further questions to narrow it down (1) • It will match symptoms to possible illnesses (1) by searching/using a database/other data store (1)		2
5(a)(ii)	Award **one** mark from: 1. May not have access to the internet (1) 2. May not have access to a device (1) 3. May not want to use it (1) 4. May not have the technical knowledge to use it (1) 5. May have a physical disability that stops them from using the service (1) 6. May not want to disclose personal information (1) 7. hacking or security or data issue (1) 8. May not trust the AI/system (1) 9. Can't take physical measurements eg. blood pressure (1) 10. may want a real person (1)		1

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Answer 3 of 5 · Question 5 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark for: • Local area network / LAN /VLAN		1
5(b)(ii)	Award **one** mark for: • Wide area network / WAN		1

Question Number	Answer	Additional Guidance	Mark
5(c)	**The only correct answer is C** *A is not correct because this is phishing* *B is not correct because this is shoulder surfing* *D is not correct because this is pharming*		1

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Answer 3 of 5 · Question 5 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award up to two marks for a linked explanation such as:

• Robust/less likely to break it if dropped / quieter than a mechanical hard drive (1) because it doesn't have any moving/mechanical parts (1)
• Smaller/thinner/lighter than a device with HDD (1) so easier to carry/more portable / laptops have less room in them for drives/componants(1)
• SSD uses less power (1) so battery will last longer (1)
• SSD is faster (1) so can access data/records more quickly (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(ii)</td><td>Any two from:

• Flash memory chips are used (1)
• Chips have (floating gate) transistors/electron pools/charge traps (1)
• Gates/electron pools/charge traps hold an electrical charge (1)
• Charge remains even when no power (1)
• No charge/empty pool/trap represents data/1 (1)
• A charge/full pool/trap represents no data/0 (1)
• Data is stored in blocks and pages (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(iii)</td><td>Award one mark for each of:

<table><tr><th>Description</th><th>RAM</th><th>ROM</th></tr><tr><td>Stores the boot up sequence</td><td></td><td>▯</td></tr><tr><td>The contents are lost when the laptop is shut down</td><td>▯</td><td></td></tr></table></td><td></td><td>2</td></tr>
<tr><td colspan="3">Total for question 5</td><td>12</td></tr>
</table>

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Answer 3 of 5 · Question 5 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

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Answer 4 of 5 · Question 1 part e · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A*** *is not correct because compression does not encrypt a file* ***B*** *is not correct because* encryption does not ensure both sides use the same key ***D*** *is not correct because* encryption does not stop hackers accessing a network		1

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Answer 4 of 5 · Question 1 part e · 2 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

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Answer 4 of 5 · Question 1 part e · 2 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A*** *is not correct because 5G does not have a* **high transmission latency** ***B*** *is not correct because a 5G does not have* ***built-in security*** ***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

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Answer 4 of 5 · Question 1 part e · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark

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Answer 5 of 5 · Question 5 part b.iii · 2 marks

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5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

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Answer 5 of 5 · Question 5 part b.iii · 2 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		**1**
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	**2**
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		**2**

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Answer 5 of 5 · Question 5 part b.iii · 2 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

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Answer 5 of 5 · Question 5 part b.iii · 2 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part b.iii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
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Answer 5 of 5 · Question 5 part b.iii · 2 marks

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Question	Answer	Additional Guidance	Mark